Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/20/2007
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Project Title:	New Automated System for Real-Time Bio-Imaging of a Vascular Cell Function in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35205-4831	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (11/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
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	The purpose of this project was to develop a novel human tissue bioimaging technology applicable to microgravity research. This system was to integrate automated bioreactor sampling and real-time bioimaging. This will enable researchers to monitor, for the first time, effects of simulated microgravity on an important vascular function--adhesion between white blood cells and the blood vessel wall-- without interruption of the culture conditions and with essentially no delay between sample extraction and cell-function assay. This has been accomplished. Technology developed in this project ultimately has potential applications for monitoring astronaut immune system function on long-term space flights. Task Objective: To develop a new, automated system for real-time bio-imaging of effects of microgravity on cell-cell adhesion through completion of the following objectives:		

Task Description:	1.Design and develop a system to automatically extract and deliver cell samples from a rotating wall vessel to an assay system at predetermined intervals. 2.Design and develop a small, integrated, real-time digital bio-imaging system to produce digital images of cell samples. 3.Integrate into the bio-imaging system an assay of vascular cell function that utilizes the unique Amgel matrix system and sensitive real-time computer tracking of adhesive cell interactions. The system developed for this purpose can be adapted to assay other effects of simulated microgravity on cultured cells. It can also be useful as a compact, portable flow-cell based adhesion assay to measure adhesion of leukocytes or other cells under shear-stress conditions similar to those found in the blood stream. Task Significance: It is critical to NASA's mission for human exploration and development of space to understand the influence of the space environment, specifically microgravity, on human physiology. Previous work has shown that microgravity can modify host cell-cell and cell-matrix interactions, and thereby exert potentially deleterious effects on human vascular and immune functions and regeneration processes. For simulated microgravity, such as in a rotating wall vessel (RWV), adhesion assays as currently conducted have important limitations. First, to remove a cell sample from the RWV, rotation must be stopped, interrupting the simulated microgravity, not just for the sample, but for all cells in the chamber. Second, adhesion assays can take over an hour to perform when one considers cell preparation, transfer of cells from the microgravity chamber to the assay, and incubation periods. While these delays may be acceptable for general adhesion research, it is important to minimize elapsed time when studying gravitational effects, so that the cells won't have time to recover from microgravity. In this project, we developed a novel human tissue bio-imaging technology applicable to microgravity research to overcome these shortcomings. Lessons learned from development of this Earth-based system can form the basis for a future space-flight culture and assay system to allow NASA to monitor microgravity-inducible physiological events during space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Adhesion of leukocytes to endothelial cells in blood vessels is a key, early step in initiation of inflammation, without which the immune response is ineffective. Much remains to be learned about this process and its molecular basis. Our efforts to better understand the effect of microgravity on leukocyte adhesion will contribute important knowledge to the field of immunology, potentially identifying new targets for therapeutic intervention into diseases involving both immune deficiency and autoimmune mechanisms.
Task Progress:	Task Progress. Objective 1: Development of the cell extraction and delivery system. Status: Completed. Even though extraction of fluid from a rotating RWV was a relatively straightforward engineering project, extraction of the cells from a rotating system is not trivial. Beyond the technical issue of a leak-free rotating interface between the chamber and external tubing, there is the issue that cells are often not uniformly distributed within the chamber. Although various models have been proposed to describe the location of cells in such a system, models do not always agree. We determined the best sample points empirically. Our current system reliably delivers cells at a uniform concentration from two separate locations (after an initial mixing period of approx. 10 min.) without interrupting rotation. In the first year, feasibility was confirmed with a largely manual system of valves and a syringe pump system that used a command line interface for each action of the pump. However, entering of individual commands made experimentation cumbersome and time consuming. The control process was improved by transitioning to a graphical control program where embedded script files are created and programmed into the syringe pump. These pre-programmed script files are executed directly from the laptop graphical control program. This software configuration for control of the syringe pump streamlines the control process and also increases the reliability of the commands. In addition, a system of several manual valves in various locations has been replaced by one central multivalve that controls the entire system. These advances represent major improvements in efficient extraction of cells, optimization of valves and control points, miniaturization, and automation of the system. Objective 2: development of the real-time bio-imaging system. Status: Completed. One of the problems with previous adhesion assay systems for cells undergoing shear stress is that a research microscope is required for imaging of the process. This necessarily requires relatively long tubing between the RWV and the assay chamber, which increases the time between rotation and assay. The research microscope also costs more than all of the other components of the system the system combined. Therefore, a completely new, low cost, custom-designed optical system was developed to reliably image cell adhesion under flow conditions in our adhesion assay system. The mini-microscope was tested extensively in real experiments, and has provided video image data of sufficient quality for the high-speed motion analysis necessary for flow-chamber cell adhesion data. In addition, we have converted from a tape-storage system to direct digitization, streamlining data analysis and improving fidelity of motion tracking. Objective 3: Integration of the bio-imaging system and the cell adhesion assay. Status: Completed. In preparation for biological experiments, a new flow chamber was developed, with an improved sealing mechanism, to provide a more portable and robust flow chamber. Although beyond the scope of the current project, these modifications are an important step in development of a spaceflight system. In order to accommodate budget cuts and shortened time frame, however, it was necessary to focus on developing a basic flow-chamber-based adhesion assay, setting aside the work on the Amgel matrix system and the real-time computer tracking. This has resulted in a complete system for assay of the microgravity effect on leukocyte adhesion that, although lacking the real-time computer analysis system originally planned, will permit state-of-the-art analysis as good as current systems in use for conventional adhesion studies. The completed system is so dependable and robust that we were able to use it in parabolic flight experiments in October of 2005. Our project was chosen along with 9 other experiments for the first NASA sponsored collaboration with Zero-G Experience, a commercial company that provides individuals the opportunity to experience microgravity. We used our system to test the role of gravity (vs. hydrodynamic flow) in bringing leukocytes into first contact with a simulated blood vessel wall, determining whether this critical early step in the immune response can occur normally during spaceflight. This project was flown again in June of 2006 on the NASA C9 parabolic flight airplane to test the effect of hematocrit on this process, and evaluate the feasibility for red cell transfusion as a countermeasure. Results of these experiments are being analyzed and prepared for publication.

Bibliography Type:	Description: (Last Updated: 04/12/2018)
Articles in Peer-reviewed Journals	Kucik DF, Rouleau RL, Smith RL, Wu X, Gupta KB. "Use of an integrated flow-chamber adhesion assay for measuring leukocyte adhesion properties in simulated and actual microgravity." Gravitational and Space Biology 2006 Aug;19(2):125-6. , Aug-2006